Cover Letter — Electrical Engineering (Archiv für Elektrotechnik, Springer Nature)

Date: October 28, 2026

To: Prof. Dr. Igor Kuzle, Editor-in-Chief

Electrical Engineering (Archiv für Elektrotechnik)

Faculty of Electrical Engineering and Computing, University of Zagreb, Croatia

Subject: Submission of Original Research Article — "PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Structural Recovery for Resilient Power Transformer Thermal Dynamics Forecasting"

Dear Prof. Dr. Kuzle and Esteemed Editorial Board,

I respectfully submit the above-titled original research manuscript for consideration as an Original Research Paper in Electrical Engineering (Archiv für Elektrotechnik). This work presents PSO-TCWavKAN, a swarm-optimized temporal convolutional wavelet Kolmogorov-Arnold network for multi-step thermal dynamics forecasting of mineral-oil-immersed power transformers, directly applicable to substation dynamic thermal rating (DTR), condition-based asset management, and thermal overrun prevention in accordance with IEC 60076-7 and IEEE Std C57.91.

Alignment with the Scope of Electrical Engineering

This manuscript addresses transformer thermal dynamics forecasting—a core topic within power engineering. Our research directly builds upon and extends the methodology published by Bezerra et al. (2024), "A forecasting methodology based on second-generation wavelets and neural networks: application to power transformer oil temperature" (Electr Eng 106(3):3017–3029, DOI: 10.1007/s00202-023-02134-5). While Bezerra et al. demonstrated the efficacy of discrete wavelets with standard neural networks, our investigation substantially advances this line of research via three orthogonal innovations:

- (1) Continuous Dyadic Morlet Wavelet Filter Banks embedded in Kolmogorov-Arnold Network edges, eliminating discrete boundary distortion and providing continuous differentiable multiscale activations that bypass discrete grid re-allocation instabilities observed in standard B-spline KANs;
- (2) Multi-Kernel Linear Subspace Containment (Proposition 1), establishing that the hypothesis-space closure contains the dual-kernel linear projection model as neural routing weights approach zero; and
- (3) Comprehensive Learning Particle Swarm Optimization (CLPSO) warm-start with Clerc constriction factor ($\chi = 0.7298$), initializing critical hyperparameters across a bounded 4-D search space prior to AdamW fine-tuning and SWA.

Summary of Scientific Contributions and Empirical Findings

Theoretical Rigor: Proposition 1 provides a strict topological closure proof establishing that the hypothesis-space closure contains the dual-kernel linear projection model as gating weights approach zero, establishing representational containment of the linear multi-kernel subspace without asserting unconditional optimization or generalization guarantees.

Empirical Benchmarking: Evaluated across 16 canonical ETT benchmark configurations (ETTh1 and ETTh2; lookback $L \in \{96, 336\}$; horizon $H \in \{96, 192, 336, 720\}$; 5 independent seeds for DLinear, NLinear, and PSO-TCWavKAN; 3 seeds for PatchTST Lite). On the volatile ETTh2 benchmark, PSO-TCWavKAN outperforms competing models in 7 of 8 lookback–horizon configurations. Across all 16 configurations, it achieves nominal

improvement over NLinear (uncorrected Wilcoxon $p = 0.0443$; Holm-corrected $p = 0.133$) and competitive parity with DLinear ($p = 0.274$; +1.4% higher mean MSE due to long-horizon dispersion).

Regime-Disaggregated Ablation: An 80-cell ablation campaign across 5 structural variants confirms that within the standard operational forecasting regime ($H \leq 336$), every module omission degrades performance (+1.63% to +4.62% Mean MSE), while transparently identifying boundary trade-offs at $H = 720$.

Operational Resilience: Controlled synthetic Gaussian noise injection ($\sigma = 0.30$) bounds operational error degradation between +1.4% and +13.0% across standard operational horizons ($H \leq 336$).

Computational Profiling: With 44.8 K parameters in the base configuration and 3.4 ms batch inference latency on NVIDIA Tesla T4, the model exhibits favorable computational economy relative to patched Transformer architectures (169.6 K params, 8.6 ms latency).

Publication Model and Open Science Declarations

In accordance with Springer Nature policies for hybrid journals, this manuscript is submitted under the traditional subscription route (no open-access publication fees; 0 APC). The author has not selected the Open Access option. All training scripts, benchmark configurations, result CSV files, and evaluation codes are openly accessible in the public GitHub repository at: https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting (MIT License). Extended per-seed logs and model checkpoints are preserved on Kaggle Cloud and available upon reasonable request from the corresponding author.

Ethical Declarations

Originality: This manuscript is original, has not been published previously, and is not under consideration elsewhere.

Competing Interests: The author declares no financial or non-financial competing interests.

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Human/Animal Rights: Not applicable. Only publicly available open-source benchmark sensor datasets were used.

Thank you sincerely for your time and consideration of this manuscript.

Respectfully yours,

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